

Angel Ollin Patricio

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SUMMARY

Incoming MIT MEng student passionate about interpretability, alignment, and theory of deep learning. Background spans distributed PyTorch for decentralized ML, LLM evals, and learning dynamics of language and vision models; most recently, mechanistic interpretability of RL-induced reward hacking in LLMs.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Julia

Deep Learning: MLPs, CNNs, Transformers, LLMs, Deep RL (TD, PG, Actor-Critic), LoRA/PEFT, Fine-Tuning, Mixed Precision, Gradient Checkpointing

Interpretability: Probing, Steering, Activation Patching

Frameworks/Libraries: PyTorch, HuggingFace (transformers, accelerate), JAX, NumPy, Pandas, PyHessian, Scikit-Learn, Jupyter, verl

Tools: Git, Bash, Linux, Vast.ai, WandB, vLLM, TensorBoard, uv

EDUCATION

Massachusetts Institute of Technology

M.Eng. in Electrical Engineering and Computer Science

Starting Spring 2027

Cambridge, MA

Massachusetts Institute of Technology

B.S. in Artificial Intelligence and Decision Making (GPA 4.4/5.0)

Sep 2022 – May 2026

Cambridge, MA

• **Relevant Coursework:** Deep Learning, Reinforcement Learning, Statistical Learning Theory, NLP, Nonlinear Optimization, Real Analysis

PUBLICATIONS

• J. Yuan, B. Le, K. Le, Y. Shi, A. Singh, R. Sharma, **A. Patricio**, R. Raskar, "SONAR Web: A Platform-Agnostic Framework for Real-Time Decentralized Learning Across Heterogeneous Edge Clients," *ACM FLEdge-AI Workshop*, 2025. **Best Presentation Award.** [\[link\]](#)

RESEARCH EXPERIENCE

A Benchmark for Spatial Understanding with Origami

Feb 2025 – Aug 2025

Undergraduate Researcher, Solar-Lezama Group

MIT CSAIL

- Extended an existing Gymnasium MDP formulation of inverse origami for language model evaluation, writing tooling to automate benchmark generation from a curated set of 25 reusable target origami stored as .obj files. Acknowledged in OrigamiBench writeup. [\[link\]](#)
- Established baseline shape-reconstruction performance by comparing MCTS, PPO, random walks, and evolutionary methods against the preliminary benchmark suite. Found strong correlation between performance drop and shape complexity in all approaches except random.

SONAR Web: Real-Time Decentralized Learning Across Heterogeneous Edge Clients

Nov 2024 – Feb 2025

Undergraduate Researcher, Camera Culture Group

MIT Media Lab

- Built cross-platform federated learning clients in Python (PyTorch) and JavaScript (TensorFlow.js). Leveraged a platform-agnostic config format and a universal JSON-based weight representation for framework-agnostic communication between heterogeneous peers.
- Contributed to a WebRTC communication layer with a central-server handshake for session join and zero server involvement thereafter. Implemented chunked streaming weight transfer to share larger models between peers, bypassing WebRTC's 16MB data channel limit.
- Validated cross-platform training and weight exchange across ring, mesh, toroidal, and Erdős-Rényi topologies with up to 12 peers, verifying gradient flow on MLP and CNN baselines over MNIST and CIFAR. Core components remain in the published system. [\[link\]](#)

Masked Language Modeling for Genomics

Apr 2024 – Aug 2024

Undergraduate Researcher, Edelman Lab

MIT IMES

- Rebuilt data pipeline for DistilBERT fine-tuning on the 30M-sample CELLxGENE census dataset, writing custom collators for the transcriptomics format including both sharding and streaming to eliminate full-dataset memory pressure. Parallelized training with HF accelerate across 8+ GPUs on a Vast.ai instance, cutting epoch times from hours to minutes.
- Built automation with WandB logging, allowing full experiment suites producing a collection of masked language modeling checkpoints, trained over vectors of concatenated phenotype labels and 500+ dimensional gene expression profiles. This formulation allowed a single model to perform classification, imputation, or both depending on the masking pattern; deployed a simple Streamlit interface for inference on user samples.

SELECTED PROJECTS

Mechanistic Interpretability Testbed for Reward Hacking in Qwen3-4B

Sep 2026 – Present

Independent Project

[\[aopatric/qwen-steering\]](#)

- Extended the reproducible RL-induced reward hacking setup from [ariahw/rl-rewardhacking](#) for Qwen3-4B into an expandable testbed for mechanistic interpretability tools to study how reward hacking emerges in language models; implemented activation caching over rollouts via PyTorch forward hooks.
- Building in stages: training linear probes on residual-stream activations to detect hacking vs. honest behavior at prompt time, and implementing activation patching to localize the responsible components and steering along probe directions to suppress hacking.

Finding an Edge of Stability for LLM Post-Training

Fall 2025; revisited 2026

Course Project, MIT 6.7960 Deep Learning

[\[aopatric/post-training-EoS\]](#)

- Built a PyTorch/HF pipeline tracking global and batch sharpness (top Hessian eigenvalues via PyHessian) while fine-tuning Pythia models (70M to 1.4B) on Alpaca with full SFT and LoRA, searching for the Edge of Stability (EoS) in LLM post-training. Initial experiments found no EoS signature.
- After finding flaws in the original measurements, rebuilt curvature tooling GPU-native in PyTorch (Hessian-vector products, batched power iteration) for a $\sim 5\times$ speedup over the CPU-bound original; validated against exact Hessians on small models, reference-library solvers, and invariant checks.
- Smoke runs up to 1.4B reproduced the null result: sharpness built up early, then slowly tapered off instead of settling at a stability threshold, and near-degenerate top eigenvalues made single-eigenvalue estimates unstable. Archived.